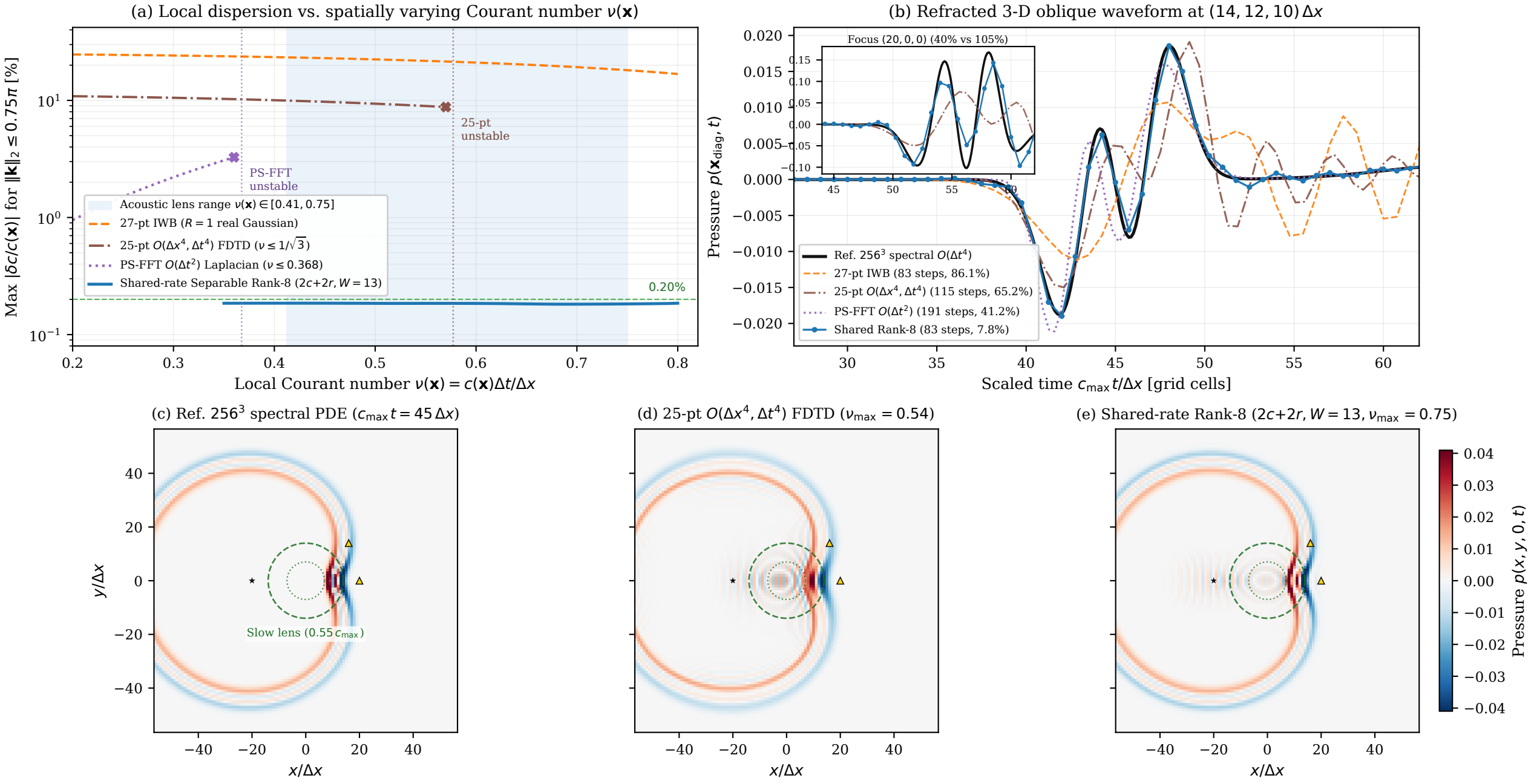

(a) Local dispersion vs. spatially varying Courant number $\nu(\mathbf{x})$
Max $|\delta c/c(\mathbf{x})|$ for $\|\mathbf{k}\|_2 \le 0.75\pi$ [%]
Local Courant number $\nu(\mathbf{x}) = c(\mathbf{x})\Delta t/\Delta x$
25-pt unstable
PS-FFT unstable
0.20%
Acoustic lens range $\nu(\mathbf{x}) \in [0.41, 0.75]$
27-pt IWB ($R = 1$ real Gaussian)
25-pt $O(\Delta x^4, \Delta t^4)$ FDTD ($\nu \le 1/\sqrt{3}$)
PS-FFT $O(\Delta t^2)$ Laplacian ($\nu \le 0.368$)
Shared-rate Separable Rank-8 ($2c+2r, W = 13$)
(b) Refracted 3-D oblique waveform at $(14, 12, 10)\,\Delta x$
Pressure $p(\mathbf{x}_{\text{diag}}, t)$
Scaled time $c_{\max} t/\Delta x$ [grid cells]
Focus (20, 0, 0) (40% vs 105%)
Ref. 256^3 spectral $O(\Delta t^4)$
27-pt IWB (83 steps, 86.1%)
25-pt $O(\Delta x^4, \Delta t^4)$ (115 steps, 65.2%)
PS-FFT $O(\Delta t^2)$ (191 steps, 41.2%)
Shared Rank-8 (83 steps, 7.8%)
(c) Ref. 256^3 spectral PDE ($c_{\max} t = 45\,\Delta x$)
Slow lens ($0.55\,c_{\max}$)
$y/\Delta x$
$x/\Delta x$
(d) 25-pt $O(\Delta x^4, \Delta t^4)$ FDTD ($\nu_{\max} = 0.54$)
(e) Shared-rate Rank-8 ($2c+2r, W = 13, \nu_{\max} = 0.75$)
Pressure $p(x, y, 0, t)$